Subtracting vectors

AS Level | 9702 AS Ch.1 | Chapter 1: Kinematics | Lesson 1.7 | Sections 1.7

Goal: Find a change in velocity, Δv = v_2 − v_1, by adding the reverse of the first vector.

velocity acceleration

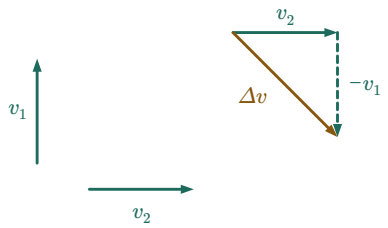
01 The big picture

To subtract a vector, add its reverse: $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$, where $-\vec{b}$ has the same size as $\vec{b}$ but points the opposite way.

The most common use is a change in velocity: $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ (final minus initial). Even at constant speed, a change of direction gives a non-zero Δv.

Change in velocity [paraphrase]
Final velocity minus initial velocity, found as a vector.

Fig. 1 · Δv = v₂ − v₁ = v₂ + (−v₁)



Draw v_2, then add the reverse of v_1 at its tip. Δv runs from the start of v_2 to the end of $-v_1$.

02 Key equations

Subtracting a vector

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$$

Change in velocity

$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

Symbols and units

$\vec{v}_1$, $\vec{v}_2$	initial and final velocity	$\mathrm{m\,s^{-1}}$	$\Delta\vec{v}$	change in velocity	$\mathrm{m\,s^{-1}}$

03 Sample problems with solutions

Sample 1 · A car turning a corner

A car travelling north at 20 m s⁻¹ turns and travels east at 20 m s⁻¹. Find its change in velocity.

Step 1. Add the reverse of v_1 (20 m s⁻¹ south) to v_2 (20 m s⁻¹ east): the two are perpendicular.

$$|\Delta v| = \sqrt{20^2 + 20^2} = 28.3\ \mathrm{m\,s^{-1}}$$

Step 2. Direction: equal east and south parts, so 45° between them.

Answer: 28 m s⁻¹ towards the south-east (45° south of east)

04 Practice problems

P1 A ball moving right at 6.0 m s⁻¹ rebounds moving left at 4.0 m s⁻¹. Find the change in velocity.
Answer: 10 m s⁻¹ to the left

P2 A velocity changes from 3.0 m s⁻¹ east to 4.0 m s⁻¹ north. Find Δv.
Answer: 5.0 m s⁻¹ at 53° north of west

P3 Find A − B for A = 7.0 N east and B = 2.0 N east.
Answer: 5.0 N east

P4 A velocity changes from 10 m s⁻¹ north to 10 m s⁻¹ south. Find Δv.
Answer: 20 m s⁻¹ south

05 Other useful information

Common mistakes

✗ **Wrong:** Subtract the speeds: 20 − 20 = 0, so no change.
✓ **Right:** The direction changed, so Δv is not zero.

✗ **Wrong:** Work out v_1 − v_2.
✓ **Right:** Δv = final − initial: v_2 − v_1.

Exam tips

- In one dimension, give each velocity a sign (+ one way, − the other) and subtract the numbers.

What the syllabus asks

- 9702 topic 1: add and subtract coplanar vectors. Coursebook 1.7.

Explore: Practice questions for this lesson